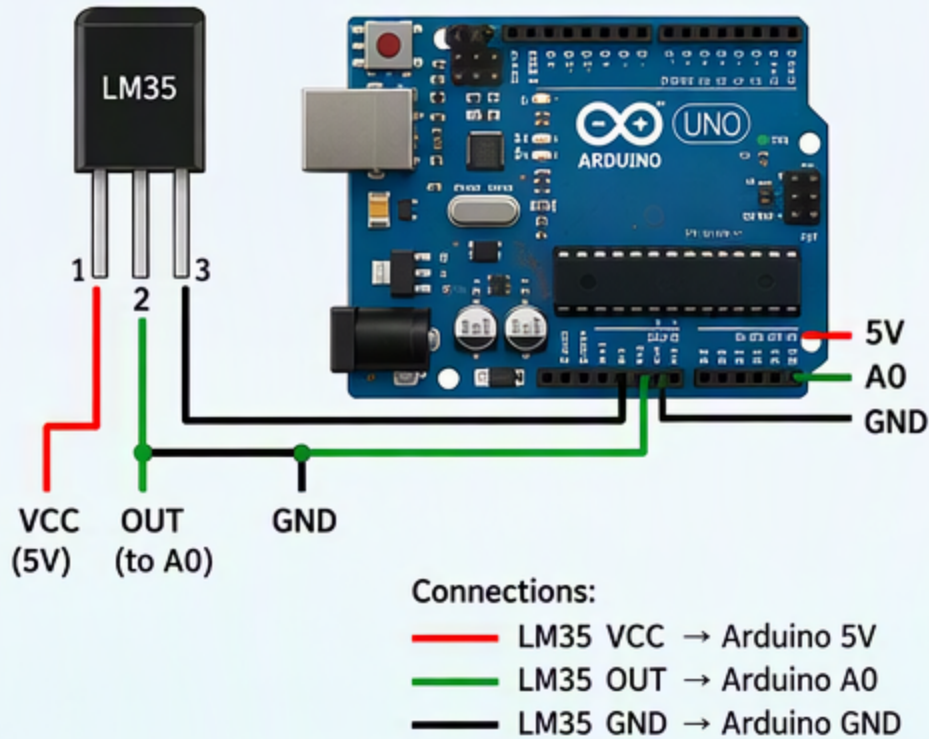




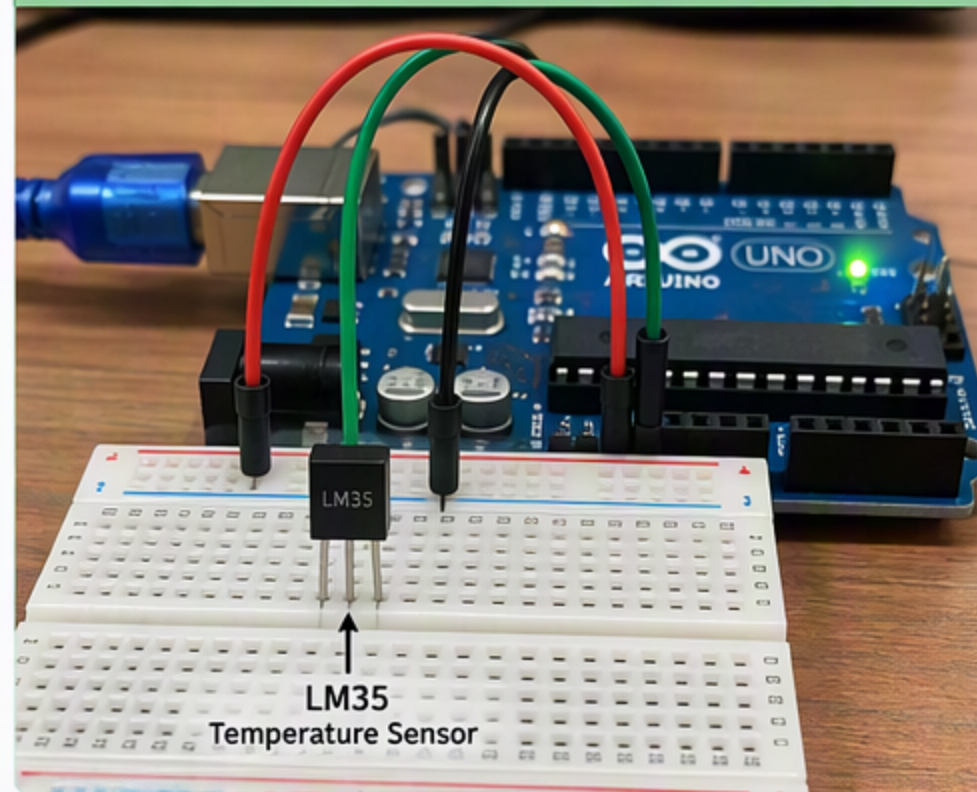
Arduino Based Temperature Monitoring System Using LM35 Temperature Sensor



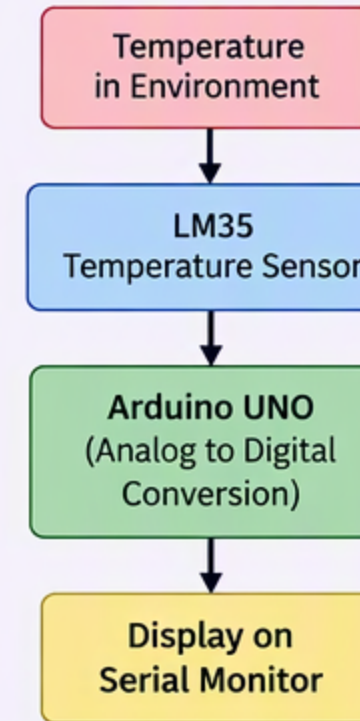
Circuit Diagram



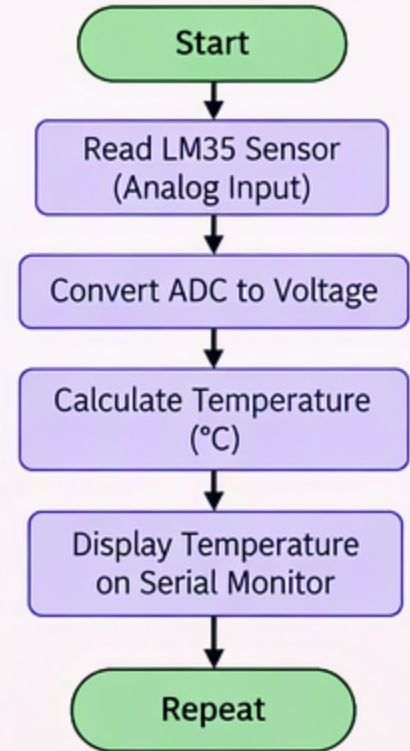
Hardware Setup (Actual Project)



Block Diagram



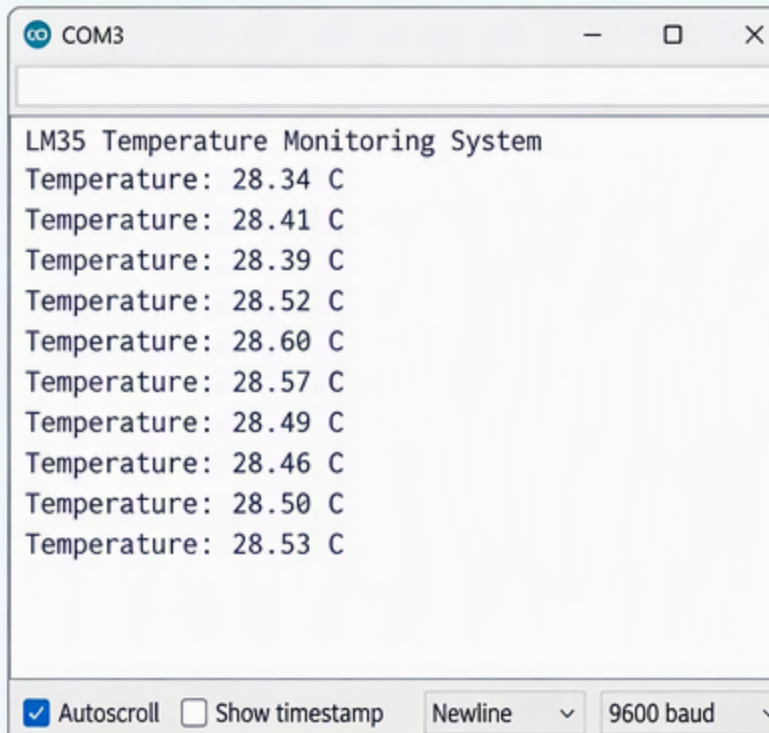
Flowchart



Arduino IDE Code

```
1 // LM35 Temperature Sensor Project
2 const int sensorPin = A0;
3
4 void setup() {
5   Serial.begin(9600);
6   Serial.println("LM35 Temperature Monitoring System");
7 }
8
9 void loop() {
10  int sensorValue = analogRead(sensorPin);
11  float voltage = sensorValue * (5.0 / 1023.0);
12  float temperatureC = voltage * 100.0;
13
14  Serial.print("Temperature: ");
15  Serial.print(temperatureC);
16  Serial.println(" C");
17  delay(1000);
18 }
```

Serial Monitor Output



Components Used

Component	Quantity
 Arduino Uno	1
 LM35 Temperature Sensor	1
 Breadboard	1
 Jumper Wires	Several
 USB Cable	1

Applications

- Room temperature monitoring
- Weather stations
- Industrial temperature measurement
- Greenhouse monitoring
- Smart home systems

Features

- ✓ Real-time temperature monitoring
- ✓ Accurate measurement
- ✓ Simple and low-cost
- ✓ Easy to implement
- ✓ Can be extended with LCD, IoT, alarms, etc.

Future Enhancements

- Add LCD display
- Add Wi-Fi (ESP8266/ESP32)
- Upload data to cloud
- Add buzzer for high temperature
- Store data on SD card
- Build a mobile app

